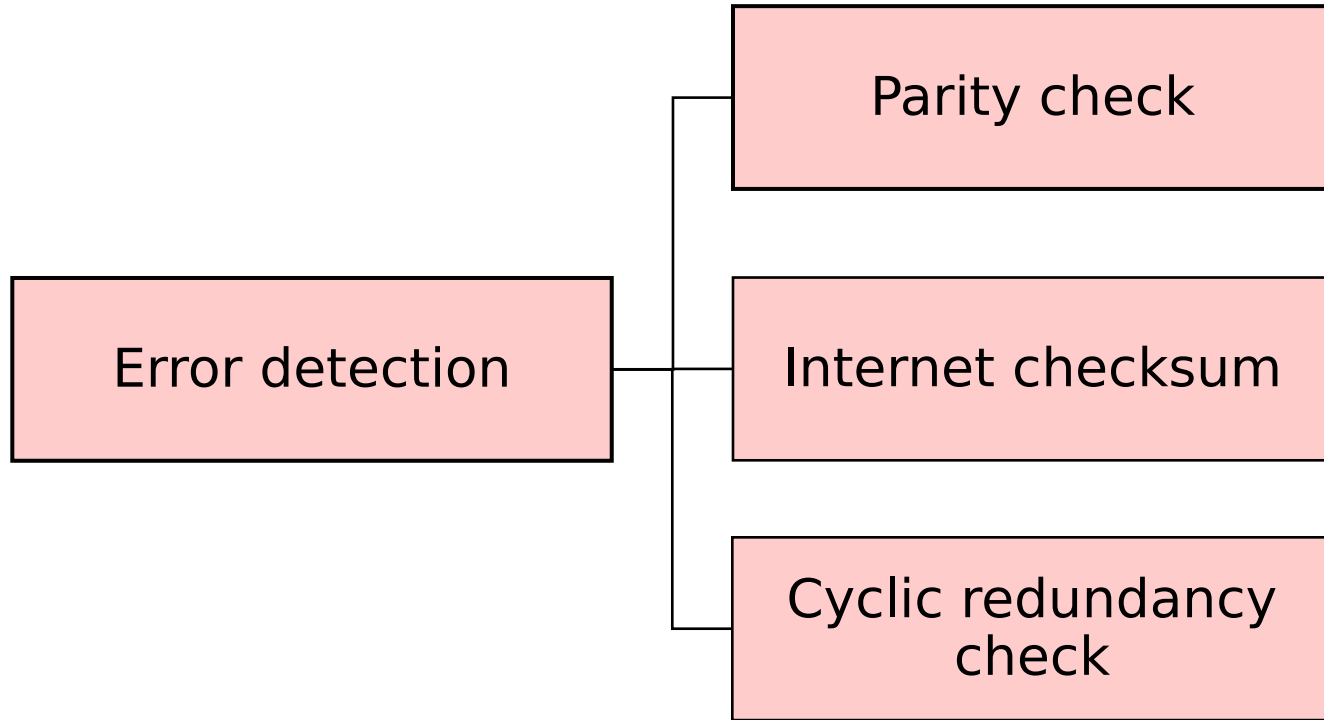


Computer Networks I

Error Detection (Cyclic Redundancy Check)

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Error Detection



Cyclic Redundancy Check

- ❑ One of the most powerful error-detecting codes
 - ❑ Two equivalent ways of presenting CRC:
 - ❑ Modulo 2 arithmetic
 - ❑ Polynomials
-

Cyclic Redundancy Check

- One of the most powerful error-detecting codes
- **Modulo 2** arithmetic:
 - Uses binary addition with no carries
 - Based on exclusive-OR (XOR) operation

$$\begin{array}{r} 1111 \\ + 1010 \\ \hline \end{array}$$

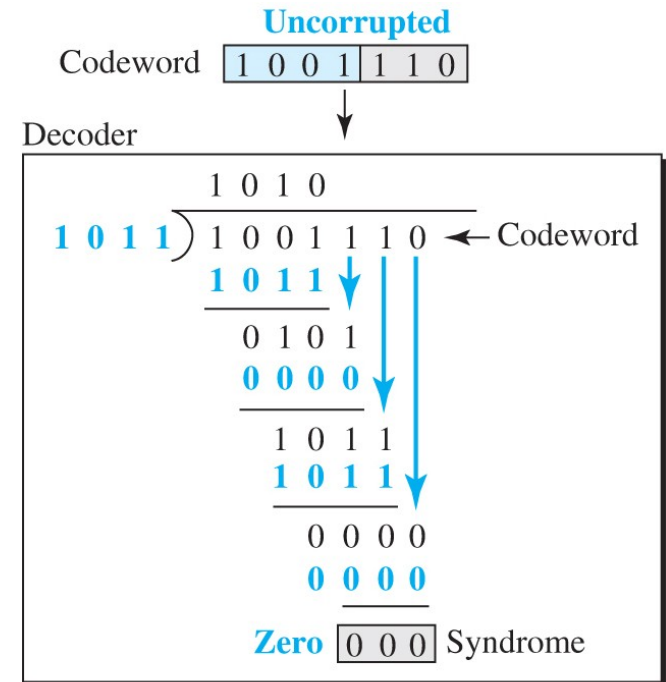
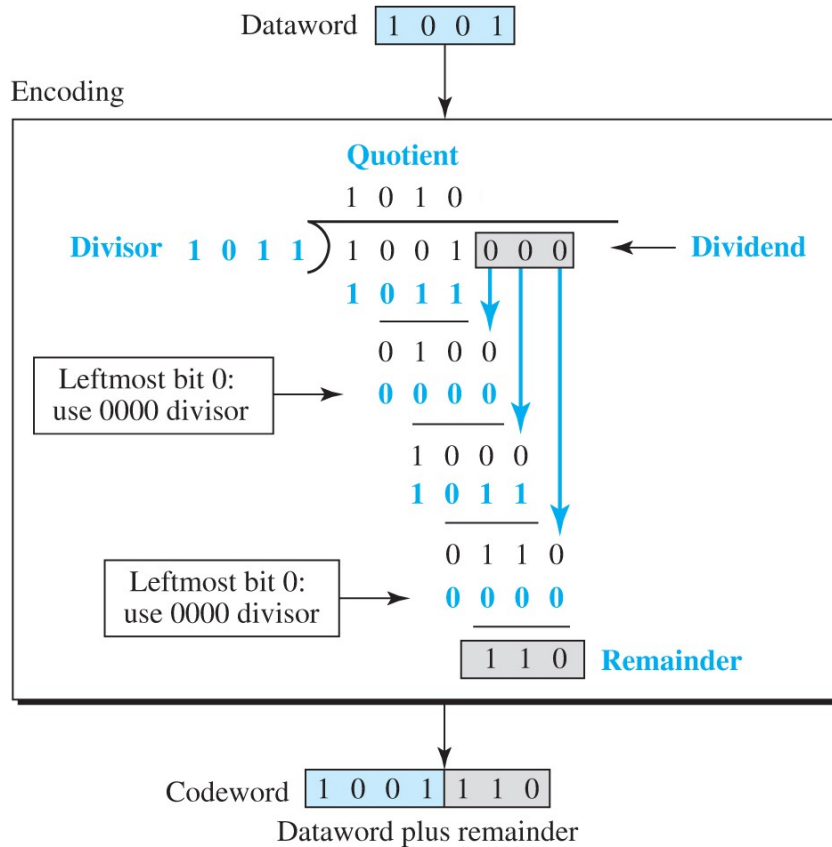
$$\begin{array}{r} 1111 \\ - 0101 \\ \hline \end{array}$$

$$\begin{array}{r} 11001 \\ \times 11 \\ \hline \end{array}$$

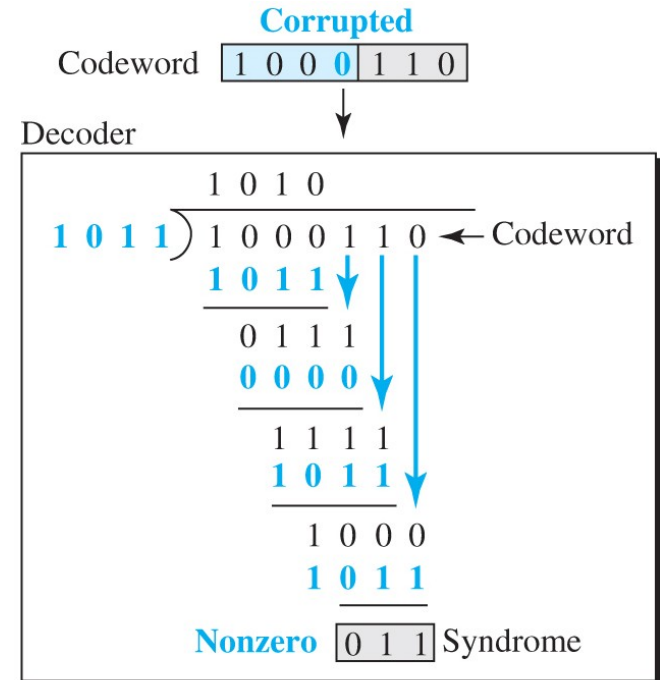
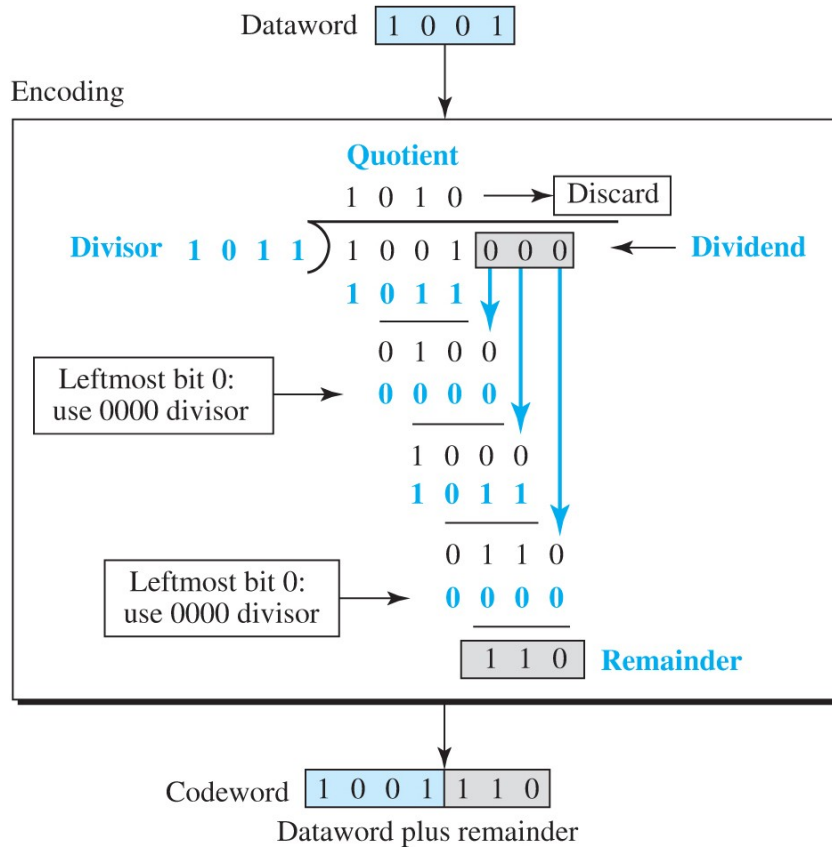
Cyclic Redundancy Check

- Given a block of bits, the transmitter generates a **frame check sequence (FCS)**, such that
 - The resulting frame is exactly divisible by some predetermined number
 - Receiver divides the incoming frame by that number
 - If there is no remainder, assume there is no error
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Cyclic Redundancy Check



Cyclic Redundancy Check



Cyclic Redundancy Check

- One of the most powerful error-detecting codes
 - Polynomials:
 - Represent bit-strings as polynomials
 - 110001 can be represented as $x^5 + x^4 + 1$
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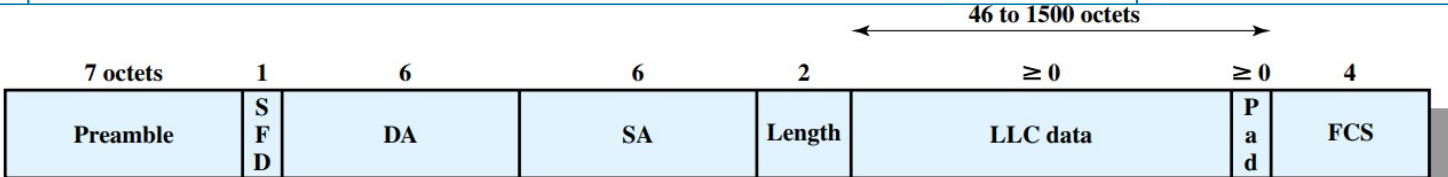
Cyclic Redundancy Check

$$D(X) = X^9 + X^7 + X^3 + X^2 + 1$$

$$P(X) = X^5 + X^4 + X^2 + 1$$

$$\begin{array}{r}
 \begin{array}{l} P(X) \rightarrow X^5 + X^4 + X^2 + 1 \end{array} \bigg/ \begin{array}{r}
 \begin{array}{r}
 X^9 + X^8 + X^6 + X^4 + X^2 + X \\
 \hline
 X^{14} \qquad X^{12} \qquad X^8 + X^7 + \qquad X^5 \\
 \hline
 X^{14} + X^{13} + \qquad X^{11} + \qquad X^9 \\
 \hline
 X^{13} + X^{12} + X^{11} + \qquad X^9 + X^8 \\
 \hline
 X^{13} + X^{12} + \qquad X^{10} + \qquad X^8 \\
 \hline
 X^{11} + X^{10} + X^9 + \qquad X^7 \\
 \hline
 X^{11} + X^{10} + \qquad X^8 + \qquad X^6 \\
 \hline
 X^9 + X^8 + X^7 + X^6 + X^5 \\
 \hline
 X^9 + X^8 + \qquad X^6 + \qquad X^4 \\
 \hline
 X^7 + \qquad X^5 + X^4 \\
 \hline
 X^7 + X^6 + \qquad X^4 + \qquad X^2 \\
 \hline
 X^6 + X^5 + \qquad X^2 \\
 \hline
 X^6 + X^5 + \qquad X^3 + \qquad X \\
 \hline
 X^3 + X^2 + X \leftarrow R(X)
 \end{array}
 \end{array}
 \begin{array}{l}
 \leftarrow Q(X) \\
 \leftarrow X^5 D(X)
 \end{array}$$

Cyclic Redundancy Check

Name	Binary	Application
CRC-8	100000111	ATM header
CRC-10	11000110101	ATM AAL
CRC-16	10001000000100001	HDLC
CRC-32	 Diagram illustrating the frame structure for CRC-32 (HDLC): - Preamble: 7 octets - SFD: 1 octet - DA: 6 octets - SA: 6 octets - Length: 2 octets - LLC data: ≥ 0 octets - Pad: ≥ 0 octets - FCS: 4 octets Total frame length: 46 to 1500 octets	

SFD = Start of frame delimiter
 DA = Destination address
 SA = Source address
 FCS = Frame check sequence

Summary

- ❑ Cyclic redundancy check
 - ❑ Modulo 2 arithmetic
 - ❑ Polynomials
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